

Adaptive Lightpath-Correlation Refinement for Efficient QoT-Aware Routing in Transparent Satellite Optical Networks

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Abstract We propose a method for efficiently routing lightpaths by adaptively switching their QoT and objective policies in transparent satellite WDM networks under stringent and time-varying transmission conditions. The method demonstrates equivalent network performance while dramatically reducing more than 70% power and 98% computation time. ©2026 The Authors

Introduction

Satellite optical networks are rapidly emerging as a cornerstone of global, high-capacity connectivity, driven by sustained advances in optical communication technologies [1, 2] and increasing satellite-internet traffic [3] and observation data [4]. Recent deployments of optical inter-satellite links (OISLs) [5] and the prospect of wavelength-division-multiplexing (WDM)-enabled space networks [6] underscore this trend. Routing and wavelength assignment (RWA) technologies for satellite WDM networks are advancing to address space-specific issues, such as the impact of delay [7], battery-aware energy efficiency [8], and dynamic topology [9]. However, dynamic quality of transmission (QoT) management remains a challenge, especially under time-varying and lossy free-space-propagation conditions.

Toward QoT-aware control in satellite WDM networks, we first introduce a QoT model that explicitly captures strong inter-lightpath interaction. This interaction stems from a satellite-specific operational principle to decouple acquisition, pointing, and tracking (APT) from RWA. Second, we resolve the computational complexity resulting from this interaction by adaptively refining the lightpath correlation only for QoT-critical OISLs. This enables scalable and power-efficient RWA without degrading performance such as blocking rate and latency.

Architecture and Operational Principle

The system architecture includes a satellite WDM network and its control system, as illustrated in

Fig. 1(a). The control core includes (i) a dynamic topology algorithm that maintains OISLs under moving satellites and (ii) RWA algorithm operating on the time-varying topology. Fig. 1(b) shows the node architecture, where the node degree is bounded by the number of laser communication terminals (LCTs) per satellite. A hierarchical switching structure combining wavelength cross-connects (WXC) and digital cross-connects (DXC) enables efficient wavelength-level switching and traffic grooming, consistent with recent satellite WDM-network designs [1, 8].

As an operational principle for flexible lightpath control, the total link power is maintained at a stable level, as the current OISL operation [10], to separate the physical topology layer from the RWA layer. Since the APT, which maintains the OISL, is controlled using the communication branch light [11], this assumption is not merely to simplify control architecture but avoid destabilization of tracking and adverse effects on dynamic topology control caused by changes in received power resulting from dynamic lightpath assignment. In addition, while the power per channel in fiber-optic networks is determined by the Kerr nonlinearity, there is no such nonlinearity in space; therefore, increasing the transmit power directly improves the signal-to-noise ratio (SNR). Consequently, this principle that adjusts channel power according to the number of WDM channels to maximize the lightpath SNR if beneficial in terms of QoT as well. However, when assigning new lightpaths, this strat-

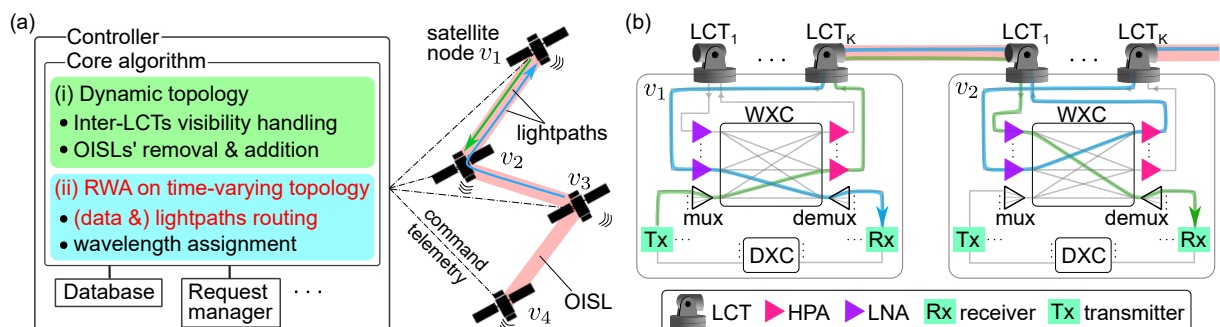


Fig. 1: Illustrative example of (a) control-system and (b) satellite-node architecture

Tab. 1: Notations and values

Item (Notation)	value	unit
Antenna diameter (ϕ)	0.1	m
HPA's noise figure (F_{HPA})[13]	6	dB
LNA's noise figure (F_{LNA})[13]	5	dB
LNA's output power (P_{LNA})	0.2	W
HPA's output power (P_{HPA})	0.5–5	W
Signal bandwidth (B) [14]	32	GHz
Receivable SNR (SNR_{min}) [15]	6	dB
Typical wavelength (λ)	1.55	μm
Pointing-error angle (θ_{err})	10^{-5}	rad
LCT's fixed-optical loss (L_{opt})	5	dB
WXC loss (L_{WXC})	15	dB

egy necessitates the challenge of considering the impact on the SNR of existing lightpaths. We believe this challenge can be overcome with the QoT model and proposed routing method.

Span-SNR and Lightpath-QoT Models

For the lightpath QoT, we define an SNR for each link e , comprising intra-node and link losses, as

$$\text{GSNR}_e^{-1} = \left[\frac{P_{\text{LNA}} L_{\text{WXC}}}{F_{\text{HPA}} h \nu B} \right]^{-1} + \left[\frac{P_{\text{HPA}} L_{\text{OISL},e}(t)}{F_{\text{LNA}} h \nu B} \right]^{-1} \quad (1)$$

where

$$L_{\text{OISL},e}(t) = L_{\text{opt}} \eta_{\text{ant}}(\theta_{\text{err}}, \phi, \lambda) \left(\frac{\pi \phi^2}{4 \lambda z_e(t)} \right)^2. \quad (2)$$

The definition and values of the terms appearing in the above equations are listed in Table 1, assuming a 100-Gbps DP-QPSK optical channel. Link loss $L_{\text{OISL},e}$ differs significantly from the SNR model for optical fiber networks; η_{ant} represents an antenna efficiency including pointing errors (~ 4 dB loss) in the far-field pattern derived from diffraction theory [12]; and the SNR varies with the time-dependent inter-satellite distance z_e .

Our QoT model is an extension of that for fiber networks [16], accounting for inter-lightpath correlation. In the base QoT model, the end-to-end QoT of a lightpath is expressed as $\mathbf{y}(t) \triangleq \mathbf{M}(t)\mathbf{x}(t)$. The notation $\mathbf{y} \in \mathbb{R}^L$ is the end-to-end vector, where y_ℓ^{-1} is the end-to-end SNR_ℓ of lightpath $\ell \in \{1, \dots, L\}$ at time t . $\mathbf{M} \triangleq (\mathbf{p}_1, \dots, \mathbf{p}_L)^T \in \{0, 1\}^{L \times |\mathcal{E}|}$ is a lightpath matrix. $\mathbf{p}_\ell(t) \in \{0, 1\}^{|\mathcal{E}(t)|}$ is the row vector of ℓ -th lightpath, the element of which is 1 if the lightpath exists on $e \in \mathcal{E}$. Here, $\mathcal{E}(t)$ is a time-varying set of OISLs. $\mathbf{x}(t) \in \mathbb{R}^{|\mathcal{E}(t)|}$ is a link-level vector, and $x_e^{-1} = \text{GSNR}_e(t)$ by Eq. (1).

Lightpath QoT is calculated by $\mathbf{y} = \mathbf{H}(t)\mathbf{x}$, given the operational principle with a strong inter-lightpath interaction. The notation $H_{\ell,e}(t) \triangleq M_{\ell,e} N_e(t)$ is a lightpath correlation matrix, where $N_e(t) \triangleq \sum_\ell M_{\ell,e}(t)$ represents the number of multiplexed wavelengths on the link e . In this model, the QoT after a new lightpath $\mathbf{p}_{L+1}$ assignment

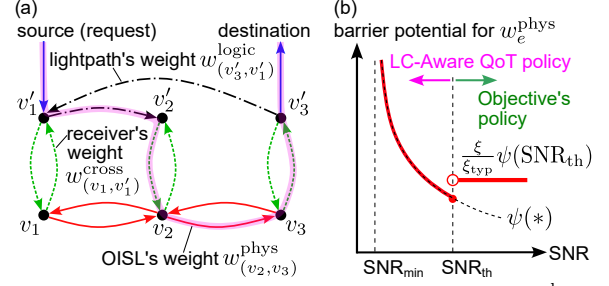


Fig. 2: (a) auxiliary graph and (b) adaptive policy for w_e^{phys}

can be calculated by

$$\mathbf{y}'(\mathbf{p}_{L+1}) \triangleq \begin{bmatrix} \mathbf{y} \\ y_{L+1} \end{bmatrix} = \mathbf{H}'\mathbf{x}, \quad (3)$$

where $\mathbf{M}' \triangleq (\mathbf{p}_1, \dots, \mathbf{p}_L, \mathbf{p}_{L+1})^T$.

Adaptive Lightpath-correlation Refinement

We developed an auxiliary-graph-based [17] routing method, which handles the lightpath correlation, for applying our QoT model to hierarchical path control with grooming capabilities. Fig. 2(a) illustrates an example where a lightpath (v_2, v_3) is newly added to assign a data path from v_1 to v_3 . The essence of the proposed routing method is to embed lightpath correlation information into the physical-layer weight $w_e^{\text{phys}}(t)$, enabling lightpath-correlation-aware (LC-aware) QoT routing. Specifically, the weight is calculated as

$$w_e^{\text{phys}} = \sum_\ell \left[M'_{\ell,e} \psi(\text{SNR}_\ell) |_{\text{SNR}_\ell \leftarrow y'_\ell(\mathbf{p}_e)^{-1}} \right], \quad (4)$$

where $\psi(\text{SNR}) \triangleq 1/(\text{SNR}/\text{SNR}_{\text{min}} - 1)$ is the barrier function and $\mathbf{p}_e$ is a row vector that equals 1 only on link e . Eq. (4) means that if a new lightpath is added to e , the barrier increases as the SNR margin of the lightpaths on e decreases.

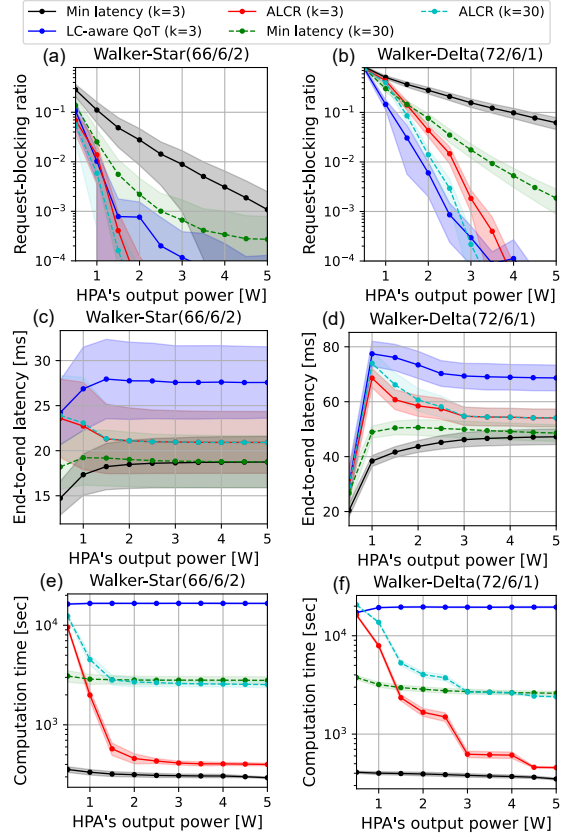
This weight function provides information on the correlation among all lightpaths for all OISLs; however, it imposes a heavy computational burden. Therefore, by combining the concept of *adaptive mesh refinement* from computational fluid dynamics [18] with a two-variable-aware-barrier approach [19, 20], we introduce an adaptive weighting scheme called *adaptive lightpath-correlation refinement* (ALCR), as Fig. 2(b) and below

$$w_e^{\text{phys}} = \begin{cases} \text{Eq.(4)} & \text{if } \frac{\text{GSNR}_e}{N_e + 1} \leq \text{SNR}_{\text{th}} \\ \frac{\xi(t)\psi(\text{SNR}_{\text{th}})}{\xi_{\text{typ}}(N_e + 1)} & \text{otherwise} \end{cases} \quad (5)$$

where SNR_{th} is the threshold for switching policies between an optimization for a variable ξ and LC-aware QoT constraints. Sensitivity analysis (not shown) suggests that SNR_{th} in the range of $[\text{SNR}_{\text{min}} + 2, \text{SNR}_{\text{min}} + 4]$ dB performs well.

Tab. 2: Routing policy and weight on auxiliary graph

Routing policy	w_*^{logic}	w_*^{phys}
Min latency	path-length	link-length
LC-aware QoT	0	Eq. (4)
ALCR (Proposed)	$\frac{\text{path-length}}{\xi_{\text{typ}}}$	Eq. (5)

**Fig. 3:** System performances with different routing policies.

Simulation Setup and Results

The performance of proposed method was evaluated by comparing the three routing policies listed in Table 2, where the policy is controlled by the weights of the auxiliary graph ($w_*^{\text{cross}} = 1$). For ALCR, ξ was set to z to minimize latency for fair comparison, and ξ_{typ} was set to the average OISL distance. Each request's paths on the auxiliary graph were calculated by the k-shortest-path algorithm. Wavelengths were assigned by first-fit.

The setup includes satellite orbits of the theoretical Walker-Delta (72/6/1); 1000km; 53° (72 satellites, 6 orbital planes, a phasing factor of 1, 1000 km altitude, and 53° orbital inclination), and the Iridium-like-practical Walker-Star (66/6/2); 780km; 86.4°, providing a sufficient diversity [21]. The Manhattan topology is maintained [22] with a time-dependent laser-topology model [23]. Traffic distribution evolves based on population distribution [24] and a gravity-traffic model [8]. Poisson-type requests are dynamically generated with an average capacity of 10 Gbps and duration of 30 s. The total-network load is set to 1 Tbps. Each node has 4 LCTs and 8 transceivers, and each link has 40 wavelength slots.

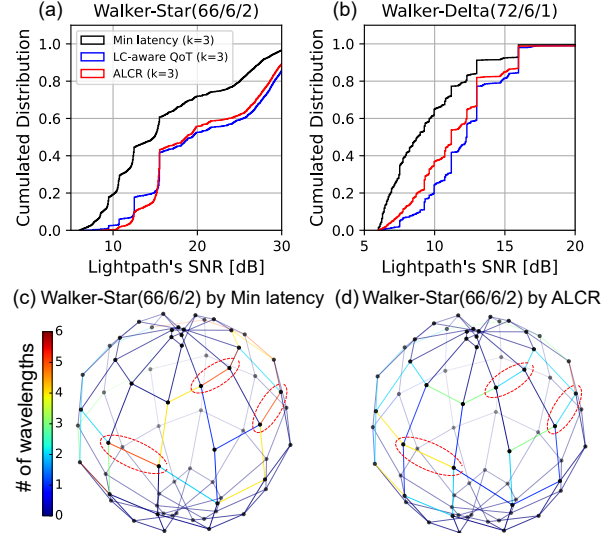
**Fig. 4:** (a–b) Distribution of end-to-end SNR of lightpaths in one trial, and (c–d) distribution of number of multiplexed wavelengths for Walker-Star with Min latency and ALCR.

Figure 3 shows the dependence of system performances on HPA output power P_{HPA} with the standard deviation (shaded area) of 32 trials with different random seeds (12-h orbit propagation for each trial). For $k = 3$ route candidates, in Fig. 3(a,b), ALCR reduced the HPA output power at a blocking rate of 10^{-3} by more than 70% compared with that of Min latency. In the effective regions of $P_{\text{HPA}} \geq 3$ W of Figs. 3(c–f), the latency was nearly equivalent to that of Min latency policy and the computation time was reduced by more than 98% compared with that of LC-aware QoT. Also, in Fig. 3(a,b), the difference in blocking rate between $k = 3$ and $k = 30$ was small for ALCR, indicating that it can efficiently approach an optimal solution with a small number of route candidates.

Figure 4 shows the detailed results for $P_{\text{HPA}} = 3$ W. In Figs. 4(a,b), ALCR improved the lightpath SNR by approximately 3–5 dB compared with Min latency, achieving nearly identical performance with LC-aware QoT. Regarding this difference between ALCR and Min latency, Figs. 4(c) and (d) show the distribution of the number of multiplexed wavelengths in the Walker-Star configuration at 5 h after the simulation started when controlled with Min latency and ALCR, respectively. When solved using Min latency, the non-uniformity of the wavelengths became relatively large, resulting in SNR bottlenecks in some areas, as shown in the area enclosed with the red dotted line in Fig. 4(c). ALCR mitigated this issue, as shown in Fig. 4(d).

Conclusion

We demonstrated that ALCR enables efficient QoT-aware routing in transparent satellite WDM networks, reducing required HPA power by over 70% and computation time by 98% without sacrificing latency or blocking performance.

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